

Practical No. 07

Aim: To study and implement different types of Constrains.

Software Requirements: MySQL, Oracle 10g.

Theory:

Constraints are rules applied on table columns to ensure **accuracy, validity, and consistency** of data in a database. They prevent the entry of invalid or inconsistent information.

1. Domain Constraints

Domain constraints restrict the values that can be stored in a column to a specific **data type, range, or format**.

- **NOT NULL** → Ensures that a column cannot be left empty.
Syntax: column_name data_type NOT NULL
- **DEFAULT** → Automatically provides a value for a column when none is supplied.
Syntax: column_name data_type DEFAULT default_value
- **CHECK** → Ensures that values entered in a column meet a specified condition.
Syntax: column_name data_type CHECK (condition)

2. Integrity Constraints

Integrity constraints are rules that maintain the **uniqueness, correctness, and identity** of data within a table.

- **PRIMARY KEY** → Uniquely identifies each row in a table (cannot be NULL or duplicate).
Syntax: column_name data_type PRIMARY KEY
- **UNIQUE** → Ensures that all values in a column are distinct (allows only one NULL).
Syntax: column_name data_type UNIQUE

3. Referential Constraints

Referential constraints enforce **relationships between tables**, ensuring that data in one table corresponds correctly to data in another.

- **FOREIGN KEY** → Ensures that a value in one table matches an existing value in another (parent) table.
Syntax: FOREIGN KEY (column_name) REFERENCES parent_table (parent_column) ON DELETE action ON UPDATE action

CASCADE → Automatically applies the parent's change to the child table.

- **ON DELETE CASCADE** → If a parent row is deleted, the matching child rows are also deleted.
- **ON UPDATE CASCADE** → If the parent's key is updated, the matching child rows update too.

Program:

```
CREATE DATABASE constraints_demo;  
USE constraints_demo;
```

```
CREATE TABLE Students (  
    roll_no INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    age INT CHECK (age >= 18),  
    city VARCHAR(50) DEFAULT 'Nagpur'  
);
```

```
INSERT INTO Students (roll_no, name, email, age, city) VALUES  
(1, 'Alice', 'alice@mail.com', 20, 'Mumbai'),  
(2, 'Bob', 'bob@mail.com', 22, 'Nagpur'),  
(3, 'Charlie', 'charlie@mail.com', 19, 'Pune');
```

```
CREATE TABLE Results (  
    result_id INT PRIMARY KEY,  
    roll_no INT,  
    marks INT,  
    FOREIGN KEY (roll_no) REFERENCES Students(roll_no) ON DELETE CASCADE ON  
UPDATE CASCADE  
);
```

```
INSERT INTO Results VALUES(101, 1, 85),(102, 2, 90),(103, 3, 78);
```

```
SELECT * FROM Students;  
SELECT * FROM Results;
```

```
INSERT INTO Students VALUES (4, 'Sam', 'sam@mail.com', 15, 'Delhi');  
INSERT INTO Students VALUES (2, 'Eve', 'eve@mail.com', 21, 'Delhi');  
INSERT INTO Results VALUES (104, 10, 75);
```

```
DELETE FROM Students WHERE roll_no = 3;  
SELECT * FROM Students;  
SELECT * FROM Results;  
DROP DATABASE constraints_demo;
```

Output:

```
mysql> CREATE TABLE Students (  
-> roll_no INT PRIMARY KEY,  
-> name VARCHAR(50) NOT NULL,  
-> email VARCHAR(100) UNIQUE,  
-> age INT CHECK (age >= 18),  
-> city VARCHAR(50) DEFAULT 'Nagpur'  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql>  
mysql> INSERT INTO Students (roll_no, name, email, age, city) VALUES  
-> (1, 'Alice', 'alice@mail.com', 20, 'Mumbai'),  
-> (2, 'Bob', 'bob@mail.com', 22, 'Nagpur'),  
-> (3, 'Charlie', 'charlie@mail.com', 19, 'Pune');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql>  
mysql> CREATE TABLE Results (  
-> result_id INT PRIMARY KEY,  
-> roll_no INT,  
-> marks INT,  
-> FOREIGN KEY (roll_no) REFERENCES Students(roll_no) ON DELETE CASCADE ON UPDATE CASCADE  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql>  
mysql> INSERT INTO Results VALUES  
-> (101, 1, 85),  
-> (102, 2, 90),  
-> (103, 3, 78);  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql>  
mysql> SELECT * FROM Students;  
+-----+-----+-----+-----+-----+  
| roll_no | name  | email          | age | city  |  
+-----+-----+-----+-----+-----+  
| 1       | Alice | alice@mail.com | 20  | Mumbai |  
| 2       | Bob   | bob@mail.com   | 22  | Nagpur |  
| 3       | Charlie | charlie@mail.com | 19  | Pune   |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql> SELECT * FROM Results;  
+-----+-----+-----+  
| result_id | roll_no | marks |  
+-----+-----+-----+  
| 101       | 1       | 85    |  
| 102       | 2       | 90    |  
| 103       | 3       | 78    |  
+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql>  
mysql> -- Domain violation: age < 18  
mysql> INSERT INTO Students VALUES (4, 'Sam', 'sam@mail.com', 15, 'Delhi');  
ERROR 3819 (HY000): Check constraint 'students_chk_1' is violated.  
mysql>  
mysql> -- Integrity violation: duplicate roll_no  
mysql> INSERT INTO Students VALUES (2, 'Eve', 'eve@mail.com', 21, 'Delhi');  
ERROR 1062 (23000): Duplicate entry '2' for key 'students.PRIMARY'  
mysql>  
mysql> -- Referential violation: roll_no not in Students  
mysql> INSERT INTO Results VALUES (104, 10, 75);  
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails  
mysql>  
mysql> DELETE FROM Students WHERE roll_no = 3;  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> SELECT * FROM Students;  
+-----+-----+-----+-----+-----+  
| roll_no | name  | email          | age | city  |  
+-----+-----+-----+-----+-----+  
| 1       | Alice | alice@mail.com | 20  | Mumbai |  
| 2       | Bob   | bob@mail.com   | 22  | Nagpur |  
+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> SELECT * FROM Results;  
+-----+-----+-----+  
| result_id | roll_no | marks |  
+-----+-----+-----+  
| 101       | 1       | 85    |  
| 102       | 2       | 90    |  
+-----+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> DROP DATABASE constraints_demo;  
Query OK, 2 rows affected (0.03 sec)
```

Conclusion:

Hence, we have successfully studied and implemented different types Constrains.